

riffle

Reflow content into new margins in the middle of a page

Manual · version 0.1.0

1 Introduction

This package reflows a stream of content into new page margins, starting exactly where the content appears and **without forcing a page break**. It is designed for documents that mix an asymmetric (book-style, wide outer margin) layout with blocks that read better at a different width.

Three functions cover the three directions:

- `column-flow` — reflows content into two columns across the symmetric width.
- `single-flow` — reflows content into a single column across the symmetric width.
- `asymmetric-flow` — reflows content from symmetric margins back into a given asymmetric layout (the reverse direction).

All of them pull footnotes out of the flow and render them at the bottom of the reflowed block, preserve footnote numbering and `refs`, lay the reflowed block flush to the bottom of the available height, and keep the new margins on the pages that follow.

2 Requirements

The functions measure the current page and position, so use them in normal document flow (they are context functions and handle this themselves). The first-line indentation of the reflowed content follows the ambient `par.first-line-indent`, so set it explicitly:

```
#set par(first-line-indent: (amount: 2em, all: true), justify: true)
```

3 Installation

The package is available on Typst Universe:

```
#import "@preview/riffle:0.1.0": column-flow, single-flow, asymmetric-flow
```

4 API reference

4.1 `column-flow`

Reflows content into symmetric two-column pages when it begins partway down an asymmetric page. The columns are widened to the symmetric content width and shifted into the symmetric position; text already on the page is untouched.

```
#column-flow(content, gutter: 4%, count: 2, footnote-entry: none)
```

- `content` (**content**): the content to reflow. Paragraph breaks are honored.
- `gutter` (**length**, **default 4%**): the gap between columns.
- `count` (**int**, **default 2**): the number of columns.
- `footnote-entry` (**none**, **dictionary**, **or function**): styling for the manually laid-out footnote entries, mirroring `footnote.entry`.

4.2 single-flow

The single-column counterpart of `column-flow`: one column spanning the whole symmetric content width.

```
#single-flow(content, footnote-entry: none)
```

- `content` (**content**): the content to reflow.
- `footnote-entry` (**none, dictionary, or function**): as in `column-flow`.

4.3 asymmetric-flow

The reverse direction: content that begins partway down a symmetric page is reflowed into a single column spanning the whole asymmetric content area given by `margin`, and the following pages use that asymmetric margin.

```
#asymmetric-flow(content, margin, footnote-entry: none)
```

- `content` (**content**): the content to reflow.
- `margin` (**auto, length, or dictionary**): the asymmetric horizontal margin to switch to, e.g. (`inside: 1cm, outside: 3cm`) or (`left: 1cm, right: 3cm`). `inside/outside` are resolved against the current page parity, like `page.margin`. Only the horizontal margins are taken from `margin`; the top and bottom margins of the current page are preserved.
- `footnote-entry` (**none, dictionary, or function**): as in `column-flow`.

5 Examples

Each example below is a complete, compilable file. Its two rendered pages are shown next to it. The body text before the reflow and the reflowed content are separated by a `#v(1em)` in the sources; this gap is cosmetic and optional. In the page images, the **blue** dashed line marks the symmetric content boundary and the **orange** dashed line the asymmetric one.

6 Two columns on an asymmetric page

```
// Example: two columns across the symmetric width on an asymmetric page.
//
// Compile to page images with:
//  typst compile -f png --ppi 144 examples/column-flow.typ "doc/images/column-flow-
{p}.png"

#import "../lib.typ": column-flow
#import "assets/guides.typ": guides

#set page(
  paper: "a5",
  margin: (inside: 1cm, outside: 2.5cm),
  numbering: "1",
  background: guides(),
)
#set text(font: ("New Computer Modern", "SimSun"), size: 9pt, lang: "zh")
#set par(first-line-indent: (amount: 2em, all: true), justify: true, leading: 0.6em)

// Body text at the normal asymmetric margins.
#lorem(45)

#v(1em)

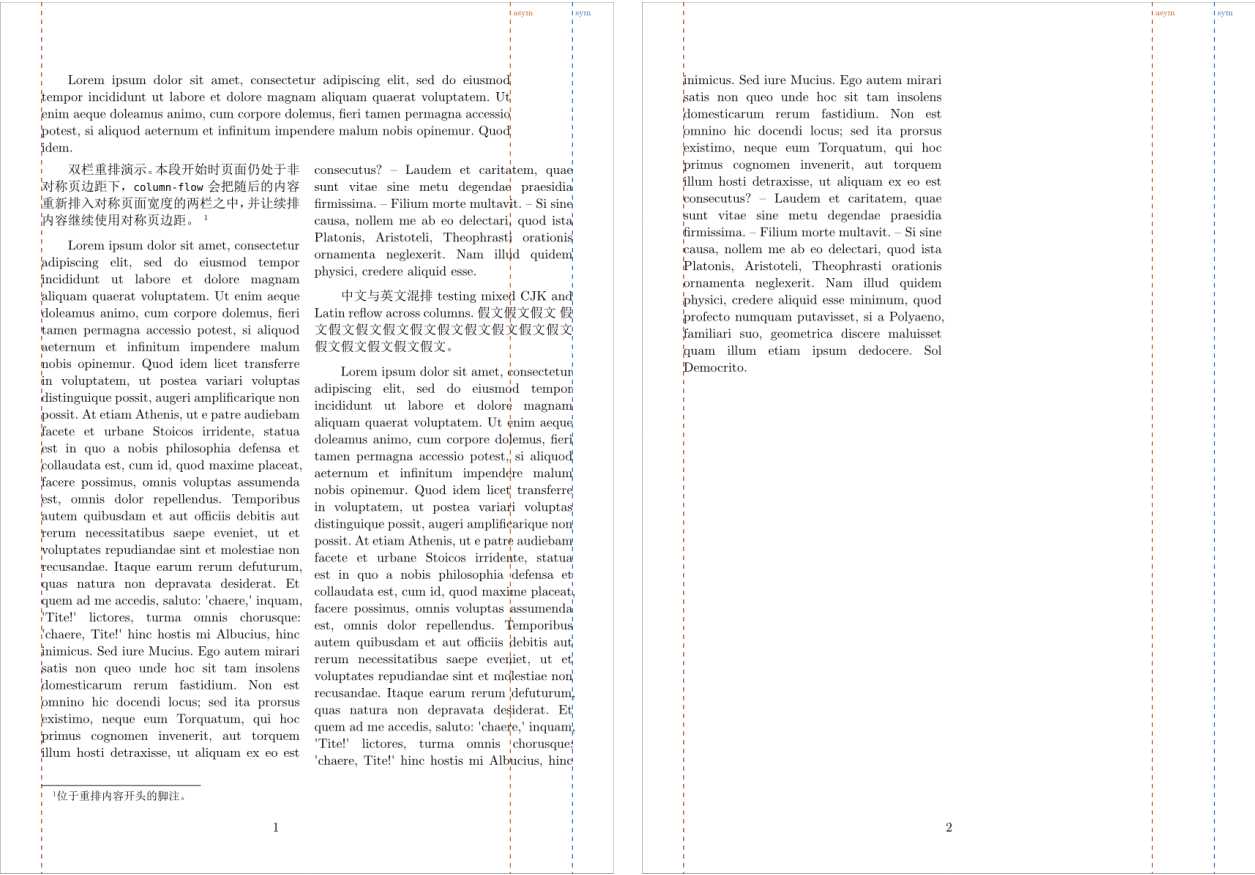
#column-flow(
```

```
[
  双栏重排演示。本段开始时页面仍处于非对称页边距下，`column-flow` 会把随后的内容
  重新排入对称页面宽度的两栏之中，并让续排内容继续使用对称页边距。
  #footnote[位于重排内容开头的脚注。]

  #lorem(230)

  中文与英文混排 testing mixed CJK and Latin reflow across columns. 假文假文假文
  假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

  #lorem(250)
],
gutter: 10pt,
)
```



Rendered output — page 1 (left) and page 2 (right). The reflowed content spans two pages, so the page-break behaviour is visible.

7 One column on an asymmetric page

```
// Example: one column across the symmetric width on an asymmetric page.
//
// Compile to page images with:
// typst compile -f png --ppi 144 examples/single-flow.typ "doc/images/single-flow-
{p}.png"

#import "../lib.typ": single-flow
#import "assets/guides.typ": guides
```

```

#set page(
  paper: "a5",
  margin: (inside: 1cm, outside: 2.5cm),
  numbering: "1",
  background: guides(),
)

#set text(font: ("New Computer Modern", "SimSun"), size: 9pt, lang: "zh")
#set par(first-line-indent: (amount: 2em, all: true), justify: true, leading: 0.6em)

// Body text at the normal asymmetric margins.
#lorem(45)

#v(1em)

#single-flow(
[
  单栏重排演示。这一段会占据对称页面的整个可用宽度，而不是像双栏那样分成两列。

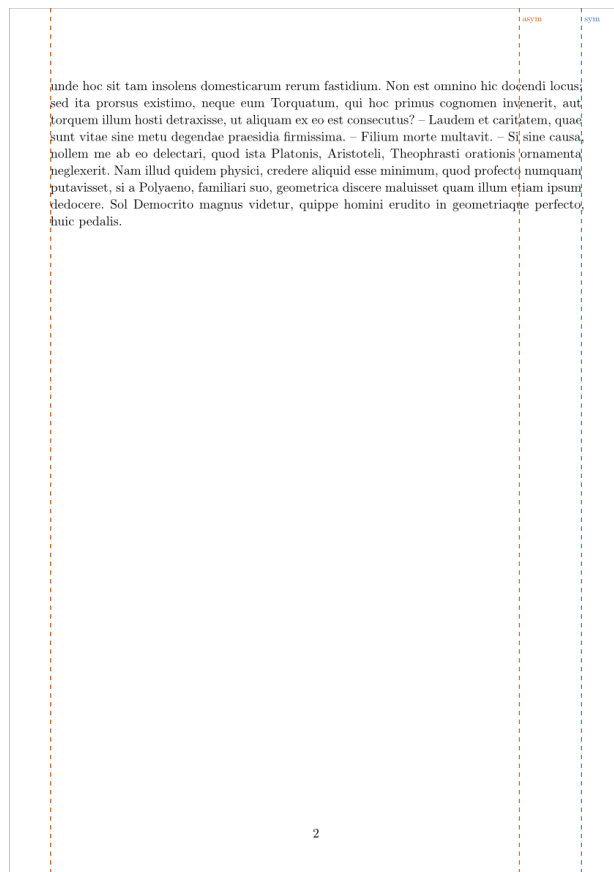
  #footnote[位于重排内容开头的脚注。]

  #lorem(240)

  中文与英文混排 testing mixed CJK and Latin reflow. 假文假文假文假文假文假文假文
  假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

  #lorem(260)
],
)

```



8 Returning to an asymmetric layout

```
// Example: switch from symmetric margins to an asymmetric layout.
//
// Compile to page images with:
// typst compile -f png --ppi 144 examples/asymmetric-flow.typ "doc/images/asymmetric-
flow-{p}.png"

#import "../lib.typ": asymmetric-flow
#import "assets/guides.typ": guides

#set page(
  paper: "a5",
  margin: 1.5cm,
  numbering: "1",
  background: guides(sym: (1.5cm, 1.5cm), asym: (1cm, 2.5cm)),
)
#set text(font: ("New Computer Modern", "SimSun"), size: 9pt, lang: "zh")
#set par(first-line-indent: (amount: 2em, all: true), justify: true, leading: 0.6em)

// Body text at symmetric margins.
#lorem(40)

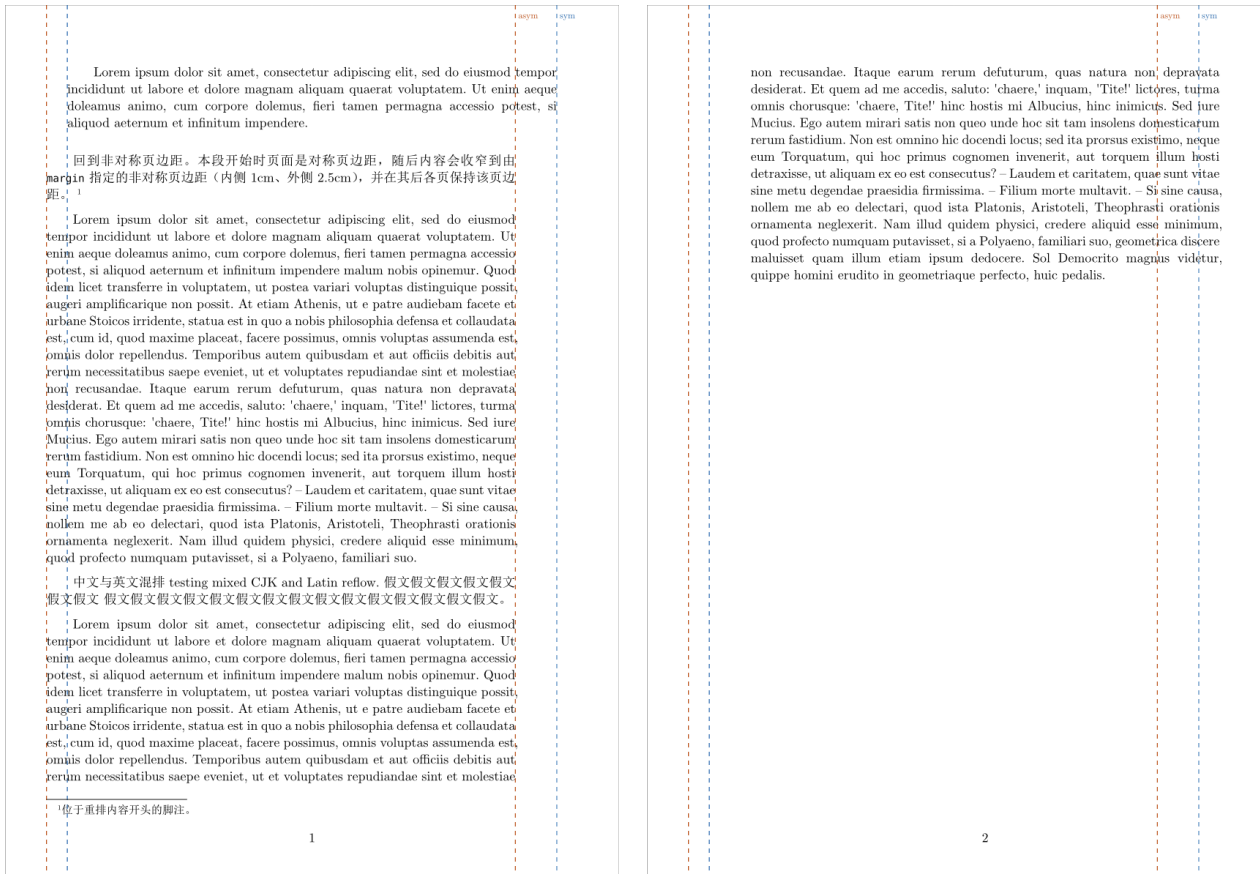
#v(1em)

#asymmetric-flow(
  [
    回到非对称页边距。本段开始时页面是对称页边距，随后内容会收窄到由 `margin`
    指定的非对称页边距（内侧 1cm、外侧 2.5cm），并在其后各页保持该页边距。
    #footnote[位于重排内容开头的脚注。]

    #lorem(240)

    中文与英文混排 testing mixed CJK and Latin reflow. 假文假文假文假文假文假文假文
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

    #lorem(260)
  ],
  (inside: 1cm, outside: 2.5cm),
)
```



Rendered output — page 1 (left) and page 2 (right). Starting from symmetric margins and switching to asymmetric ones.

9 Footnotes and a figure

```
// Example: complex content – footnotes at the start and in the middle, and an
// image, all inside the reflowed block.
//
// Compile to page images with:
//   typst compile -f png --ppi 144 examples/footnotes-figure.typ "doc/images/footnotes-
//   figure-{p}.png"

#import "../lib.typ": column-flow
#import "assets/guides.typ": guides

#set page(
  paper: "a5",
  margin: (inside: 1cm, outside: 2.5cm),
  numbering: "1",
  background: guides(),
)
#set text(font: ("New Computer Modern", "SimSun"), size: 9pt, lang: "zh")
#set par(first-line-indent: (amount: 2em, all: true), justify: true, leading: 0.6em)
#set figure(gap: 0.5em)

// Body text at the normal asymmetric margins.
#lorem(40)

#v(1em)

#column-flow(
```

```
[
  #footnote[位于重排内容最开头的脚注。] 第一段在重排内容的最开头就带有一个脚注。

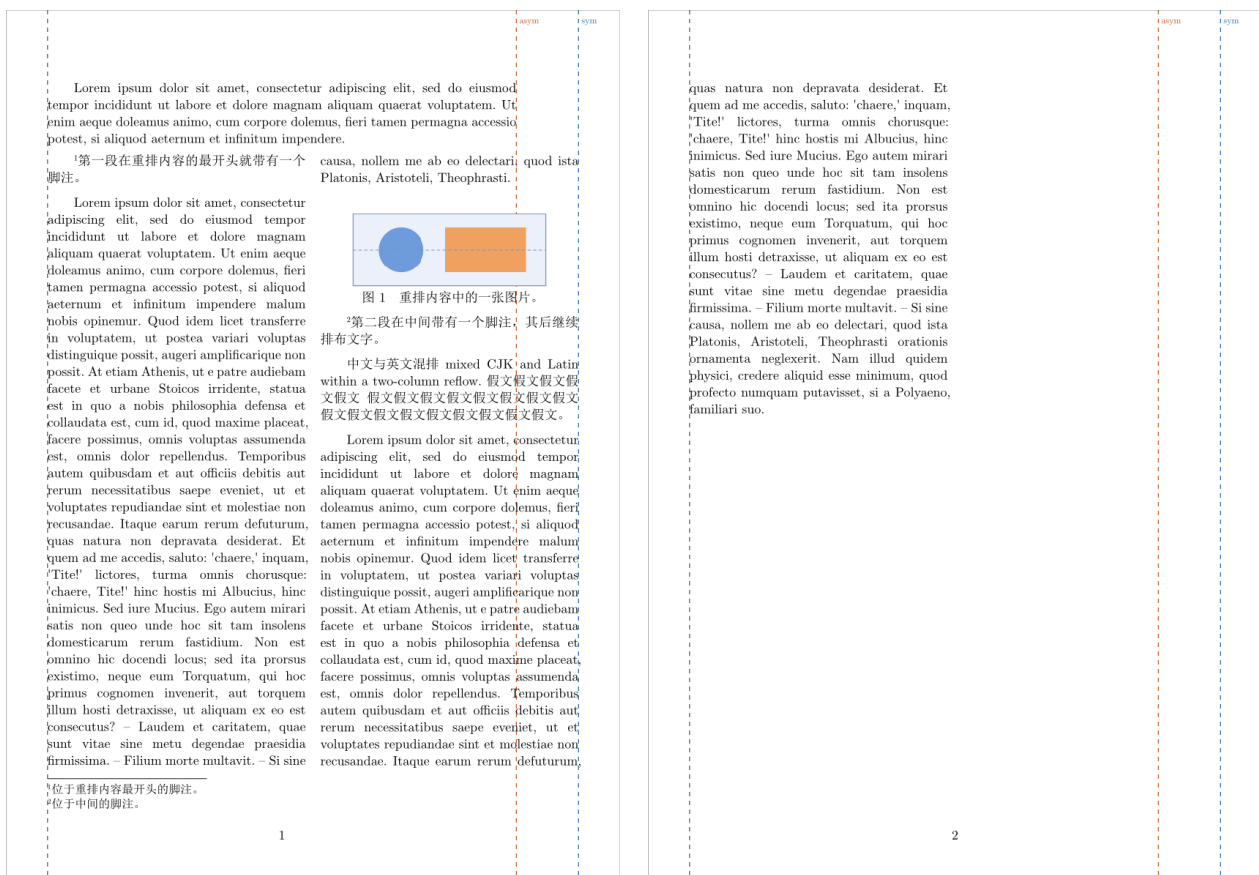
  #lorem(220)

  #figure(image("assets/plot.svg", width: 75%), caption: [重排内容中的一张图片。])

  #footnote[位于中间的脚注。] 第二段在中间带有一个脚注，其后继续排布文字。

  中文与英文混排 mixed CJK and Latin within a two-column reflow. 假文假文假文假文假文
  假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

  #lorem(240)
],
gutter: 10pt,
footnote-entry: (indent: 0em, size: 8pt),
)
```



Rendered output — page 1 (left) and page 2 (right). A footnote at the very start, a figure in the middle, and a second footnote, all inside the reflowed block.

10 Compatibility and CJK

10.1 Content-processing packages

The functions `split` and `rebuild` content, so packages that **transform content** should be applied to the content **before** it is passed in. For example, with `cjk-unbreak`:

```
#import "@preview/cjk-unbreak:0.2.3": remove-cjk-break-space

#column-flow(remove-cjk-break-space[
  中文与英文混排 content ...
])
```

10.2 CJK and mixed scripts

The splitter is aware of Han characters, CJK punctuation and spaces, so it handles Chinese, Japanese and Korean text as well as mixed CJK/Latin reflow. For best results, set a CJK-capable font and enable justification and first-line indentation:

```
#set text(font: ("New Computer Modern", "SimSun"), lang: "zh")
#set par(first-line-indent: (amount: 2em, all: true), justify: true)
```

10.3 Layout packages

The functions build their own fixed-height grid/block structure and change `page.margin`, so packages that also control pagination, columns or page geometry may conflict. Packages that only **decorate** content (colours, boxes, emphasis) generally work; layout packages have not been tested. Footnote styling is rendered manually, so use the `footnote-entry` parameter if a package restyles `footnote.entry`.

10.4 Margin-note packages (e.g. `marginalia`)

Using this package together with a margin-note package such as `marginalia` is a common case, but because the flow functions manage the page and change `page.margin` mid-page, a few precautions are needed:

- Do **not** wrap the output of a flow function in `marginalia.wideblock`. A wideblock computes its width from the `marginalia` configuration rather than from the current page margins, and a multi-page wideblock does not work with alternating margins.
- Do **not** use `marginalia.wideblock` or `marginalia.header` for headers and footers. Instead, align the header/footer directly from `page.margin` (a broader alignment that follows the margins this package sets). A minimal example adapted from the book template:

```
#set page(header: context {
  let m = page.margin
  let pad-fn = if "inside" in m {
    if calc.even(here().page()) {
      pad.with(left: m.inside - m.outside, rest: 0pt)
    } else {
      pad.with(right: m.inside - m.outside, rest: 0pt)
    }
  } else {
    it => it
  }
  pad-fn(align(end + horizon)[#counter(page).display()])
})
```

- An **empty** `marginalia.wideblock` may still be used to reserve the widened area so that margin notes do not overflow into the body text.

11 Inspiration

The idea of reflowing content mid-page comes from [meander.typ](https://github.com/Vanille-N/meander.typ). This package focuses on the asymmetric $\rightleftharpoons$ symmetric reflow direction and, in particular,

on the multi-page asymmetric case that could previously fail to [converge](<https://github.com/Vanille-N/meander.typ/issues/1#issuecomment-3306100761>). It also supports mixed CJK and Latin reflow.

The name **rifle** stays with the river metaphor: a riffle is the short stretch of a stream where the current changes character — here, the stretch of a page where the layout changes course.

12 License

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